FREE STRONG UNIMODALITY AND A CONJECTURE OF HASEBE AND UEDA

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Abstract. A probability measure μ on $\mathbb{R}$ is freely strongly unimodal if $\mu \boxplus \nu$ is unimodal for every unimodal probability measure ν. We prove that only point masses have this property. This answers Problem 3.11 of Hasebe and Ueda, who proved it for measures of finite variance. The witnesses are explicit and do not depend on μ: if μ is not a point mass, then $\mu \boxplus \left(\frac{1}{100}\delta_0 + \frac{99}{100}U[0,L]\right)$ is not unimodal for all large L. We also show that the classical half of their Conjecture 5.5 fails for one-sided stable kernels. If $0 < \alpha < 1$ and S_t is the law at time t of a positive strictly α-stable process, then for every nonnegative Borel function f there is a probability measure μ with $\int f\,d\mu < \infty$ such that $\mu * S_t$ is unimodal for no $t > 0$. In particular, a finite moment of order $2+\alpha$, or of every order, does not give unimodality in large time.

1. Introduction

Following [5], a Borel measure μ on $\mathbb{R}$ is *unimodal* if there exist $a \in \mathbb{R}$ and a function $h\colon \mathbb{R} \to [0,\infty)$ which is nondecreasing on $(-\infty, a)$ and nonincreasing on (a,∞) such that

$$\mu(dx) = \mu(\{a\})\,\delta_a(dx) + h(x)\,dx. \tag{1.1}$$

Any such a is called a mode. Translations and dilations by positive factors preserve unimodality. A probability measure μ is strongly unimodal if $\mu * \nu$ is unimodal for every unimodal ν. By Ibragimov's theorem [6], a probability measure that is not a point mass is strongly unimodal if and only if it is absolutely continuous with a log-concave density; see also [7, Theorem 52.3]. Normal and exponential distributions are therefore strongly unimodal.

Hasebe and Ueda [5, Definition 3.8] call a probability measure μ on $\mathbb{R}$ *freely strongly unimodal* if $\mu \boxplus \nu$ is unimodal for every unimodal probability measure ν on $\mathbb{R}$, where $\boxplus$ is free additive convolution. Point masses have this property, since $\delta_a \boxplus \nu$ is the translate of ν by a. Hasebe and Ueda proved that the semicircle laws do not have it, and more generally that no probability measure with finite variance other than a point mass has it [5, Theorem 3.10]. Their proof passes through the free central limit theorem and so requires a finite second moment. They asked [5, Problem 3.11] whether a freely strongly unimodal probability measure that is not a point mass exists. The answer is negative.

For $L > 0$ let $U[0,L]$ denote the uniform distribution on $[0,L]$ and put

$$\nu_L = \tfrac{1}{100}\,\delta_0 + \tfrac{99}{100}\,U[0,L].$$

Each ν_L is unimodal with mode 0.

Theorem 1.1. *Let μ be a Borel probability measure on $\mathbb{R}$ that is not a point mass. There exists $L_0 = L_0(\mu) < \infty$ such that $\mu \boxplus \nu_L$ is not unimodal for every $L > L_0$. Consequently the freely strongly unimodal probability measures on $\mathbb{R}$ are exactly the point masses.*

No moment condition is imposed on μ. After a dilation by $\varepsilon = 1/L$, the measure μ becomes close to δ_0 while ν_L becomes the fixed measure ν_1. The subordination function of the convolution then moves the Cauchy transform of ν_1 by an analytic function with values in the lower half-plane. By the Schwarz–Pick lemma, this function and all its derivatives are

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controlled near two chosen points by a single small parameter t_ε. A first-order expansion in t_ε gives the sign of the derivative of the density at two points, and the two signs are incompatible with unimodality. The tails of μ enter only through t_ε, and the estimates are relative to t_ε.

The second result concerns large-time unimodality. For classical Lévy processes it can fail: Wolfe [8] gave a class of Lévy processes whose marginal laws are not unimodal for large t. In the free setting Hasebe and Sakuma [4] proved that every free Lévy process whose Lévy measure has bounded support is unimodal for all sufficiently large t, and that the boundedness cannot be dropped. In [5, Section 5.2], Hasebe and Ueda consider the positive stable process of index $1/2$, whose law at time t is the Lévy distribution

$$L_t(dx) = \frac{t}{\sqrt{2\pi}}\, x^{-3/2} e^{-t^2/(2x)}\, \mathbf{1}_{(0,\infty)}(x)\, dx.$$

They prove that $\mu * L_t$ is unimodal for all large t when μ has compact support [5, Theorem 5.3]. They also construct a probability measure μ with $\int |x|^{5/2}\, d\mu = \infty$ for which $\mu * L_t$ is unimodal for no $t > 0$, and conjecture that finiteness of $\int |x|^{5/2}\, d\mu$ gives unimodality in large time. More generally, they state the following.

Conjecture ([5, Conjecture 5.5]). *Suppose that S_t (resp. T_t) is the law at time $t \geq 0$ of a classical (resp. free) strictly stable process of index $\alpha \in (0,2)$. If μ is a probability measure such that $\int_{\mathbb{R}} |x|^{2+\alpha}\, d\mu(x) < \infty$, then $\mu * S_t$ (resp. $\mu \boxplus T_t$) is unimodal for sufficiently large $t > 0$.*

A positive strictly α-stable process, $0 < \alpha < 1$, is a strictly stable process whose law at time t is the probability measure S_t on $[0,\infty)$ with

$$\int_{[0,\infty)} e^{-\lambda x}\, S_t(dx) = \exp(-t\gamma\lambda^\alpha), \qquad \lambda \geq 0, \tag{1.2}$$

for some $\gamma > 0$. For $\alpha = 1/2$ and $\gamma = \sqrt{2}$ this is the process L_t above.

Theorem 1.2. *Let $0 < \alpha < 1$, let $(S_t)_{t>0}$ be as in (1.2), and let $f\colon \mathbb{R} \to [0,\infty)$ be a Borel function. There is a Borel probability measure μ on $\mathbb{R}$ with $\int f\, d\mu < \infty$ such that $\mu * S_t$ is not unimodal for any $t > 0$.*

The measures in the proof are supported on $\{-2^n : n \geq 1\}$, and the weights can be any positive numbers (Proposition 4.2). Very small weights give the following.

Corollary 1.3. *Let $Z = \sum_{n\geq 1} 2^{-n^2}$ and $\mu_* = Z^{-1} \sum_{n\geq 1} 2^{-n^2} \delta_{-2^n}$. Then $\int |x|^r\, d\mu_* < \infty$ for every $r > 0$, and for every $\alpha \in (0,1)$ and every $t > 0$ the measure $\mu_* * S_t$ is not unimodal. In particular the classical half of Conjecture 5.5 of* [5] *fails for every $\alpha \in (0,1)$, and $\mu_* * L_t$ is unimodal for no $t > 0$ although $\int |x|^{5/2}\, d\mu_* < \infty$.*

In the construction of [5, Section 5.2] the atoms move to $+\infty$, in the direction of the support of the kernel. A point then receives mass from the atoms on its left through both the increasing and the decreasing part of the kernel, and the condition on positions and weights that makes the construction work forces $\int |x|^{5/2}\, d\mu = \infty$ [5, (5.45)]. Here the atoms move to $-\infty$. Between two consecutive atoms, far from the origin, only the atoms on the left contribute, and each contributes through the decreasing part of the kernel. The weights play no role.

Theorem 1.2 says nothing about symmetric kernels or about $\alpha \in [1,2)$, where every nondegenerate strictly stable law has support $\mathbb{R}$, and nothing about the free half of the conjecture. The proof uses that the classical convolution of a kernel with a discrete measure is a mixture of translates of the kernel; free convolution has no such description.

2. Preliminaries

Write $\mathbb{C}^+$ and $\mathbb{C}^-$ for the open upper and lower half-planes. For a Borel probability measure η on $\mathbb{R}$ let

$$G_\eta(z) = \int_{\mathbb{R}} \frac{\eta(ds)}{z - s}, \qquad F_\eta(z) = \frac{1}{G_\eta(z)}, \qquad z \in \mathbb{C}^+.$$

Since $\operatorname{Im} G_\eta(z) = -\operatorname{Im} z \int |z-s|^{-2}\,\eta(ds) < 0$, the function F_η is analytic on $\mathbb{C}^+$. For $c > 0$ let $D_c\eta$ be the image of η under $x \mapsto cx$. Then $D_c(\eta \boxplus \kappa) = D_c\eta \boxplus D_c\kappa$, and translating η by $a \in \mathbb{R}$ translates $\eta \boxplus \kappa$ by a. If η' is the translate of η by $-a$, then $F_{\eta'}(z) = F_\eta(z+a)$.

Lemma 2.1. *For every $z \in \mathbb{C}^+$, $\operatorname{Im} F_\eta(z) \ge \operatorname{Im} z$, and the inequality is strict for every $z \in \mathbb{C}^+$ when η is not a point mass.*

Proof. We have $\operatorname{Im} F_\eta(z) = -\operatorname{Im} G_\eta(z)/|G_\eta(z)|^2 = \operatorname{Im} z \int |z-s|^{-2}\,\eta(ds)/|G_\eta(z)|^2$. By the Cauchy–Schwarz inequality, $|G_\eta(z)|^2 \le \int |z-s|^{-2}\,\eta(ds)$, with equality only if $s \mapsto (z-s)^{-1}$ is η-almost surely constant, that is, only if η is a point mass. □

We use the analytic subordination of free additive convolution, proved for arbitrary Borel probability measures on $\mathbb{R}$ by Biane [3]; see also Belinschi and Bercovici [1]. Free convolution of measures with unbounded support is defined in [2].

Theorem 2.2 ([3, 1])**.** *Let λ, κ be Borel probability measures on $\mathbb{R}$ and $\sigma = \lambda \boxplus \kappa$. There are analytic maps $\omega_\lambda, \omega_\kappa \colon \mathbb{C}^+ \to \mathbb{C}^+$ such that, for $z \in \mathbb{C}^+$,*

$$F_\lambda(\omega_\lambda(z)) = F_\kappa(\omega_\kappa(z)) = F_\sigma(z), \qquad \omega_\lambda(z) + \omega_\kappa(z) = z + F_\sigma(z).$$

The subordination equations determine $F_\sigma(z)$ at each point.

Lemma 2.3. *Let λ, κ, σ be as in Theorem 2.2 and $z \in \mathbb{C}^+$. Suppose $\omega_1, \omega_2 \in \mathbb{C}^+$ and $u \in \mathbb{C}$ satisfy $F_\lambda(\omega_1) = F_\kappa(\omega_2) = u$ and $\omega_1 + \omega_2 = z + u$. Then $u = F_\sigma(z)$.*

Proof. Put $h_\eta = F_\eta - \mathrm{id}$, so that $\operatorname{Im} h_\eta \ge 0$ on $\mathbb{C}^+$ by Lemma 2.1. The hypotheses give $\omega_1 = z + u - \omega_2 = z + h_\kappa(\omega_2)$ and $\omega_2 = z + u - \omega_1 = z + h_\lambda(\omega_1)$. Hence ω_2 is a fixed point of

$$T_z(v) = z + h_\lambda\big(z + h_\kappa(v)\big).$$

The map T_z is analytic on $\mathbb{C}^+$ and $\operatorname{Im} T_z(v) \ge \operatorname{Im} z$ for all $v \in \mathbb{C}^+$, so T_z maps $\mathbb{C}^+$ into itself and is not the identity. After a conformal map of $\mathbb{C}^+$ onto the unit disk sending a fixed point to 0, the Schwarz lemma shows that a holomorphic self-map of the disk with two distinct fixed points is the identity. So T_z has at most one fixed point in $\mathbb{C}^+$. The pair $(\omega_\lambda(z), \omega_\kappa(z))$ satisfies the same equations with u replaced by $F_\sigma(z)$, so $\omega_\kappa(z)$ is also a fixed point of T_z. Therefore $\omega_2 = \omega_\kappa(z)$ and $u = F_\kappa(\omega_2) = F_\sigma(z)$. □

Lemma 2.4. *Let I be an open interval, $h \colon I \to \mathbb{R}$ nondecreasing, and $f \colon I \to \mathbb{R}$ continuous with $h = f$ almost everywhere on I. Then f is nondecreasing on I. The same holds with nonincreasing in place of nondecreasing.*

Proof. Let $u < v$ in I. Since the set where $h = f$ has full measure, it is dense, and we may choose $u_k \to u$ and $v_k \to v$ in this set with $u_k < v_k$. Then $f(u_k) = h(u_k) \le h(v_k) = f(v_k)$, and letting $k \to \infty$ gives $f(u) \le f(v)$. □

We also use the Stieltjes inversion formula in the following form. If G_σ extends continuously from $\mathbb{C}^+$ to $\mathbb{C}^+ \cup J$ for an open interval $J \subset \mathbb{R}$, then the restriction of σ to J is absolutely continuous with density $-\pi^{-1} \operatorname{Im} G_\sigma(x)$, $x \in J$.

3. Proof of Theorem 1.1

Throughout, μ is a Borel probability measure on $\mathbb{R}$ which is not a point mass, and $\lambda_\varepsilon = D_\varepsilon \mu$ for $\varepsilon > 0$. None of the λ_ε is a point mass.

3.1. **The measure κ.** Let $p = 1/100$, $q = 99/100$ and $\kappa = \nu_1 = p\,\delta_0 + q\,U[0,1]$. For $z \in \mathbb{C}^+$,

$$G_\kappa(z) = \frac{p}{z} + q \int_0^1 \frac{ds}{z-s} = \frac{p}{z} + q\big(\operatorname{Log} z - \operatorname{Log}(z-1)\big),$$

where Log is the principal branch. Let N be an open disk containing the segment $[1/200, 1/2]$ and contained in $\{0 < \operatorname{Re} z < 1\}$. On N define

$$A(z) = \frac{p}{z} + q\big(\operatorname{Log} z - \operatorname{Log}(1-z) - i\pi\big).$$

For $z \in N \cap \mathbb{C}^+$ we have $1 - z \in \mathbb{C}^-$ and $\operatorname{Log}(z-1) = \operatorname{Log}(1-z) + i\pi$, so $A = G_\kappa$ on $N \cap \mathbb{C}^+$. For real $x \in N$,

$$A(x) = g(x) - ic, \qquad g(x) = \frac{p}{x} + q \log \frac{x}{1-x}, \qquad c = \pi q, \tag{3.1}$$

and $A'(x) = g'(x)$ is real. A direct computation gives

$$g'(x) = \frac{x-p}{x^2(1-x)}, \qquad g''(x) = \frac{2p}{x^3} + q\Big(\frac{1}{(1-x)^2} - \frac{1}{x^2}\Big). \tag{3.2}$$

Let $x_1 = p$ and $x_2 = 1/4$. Then

$$g'(x_1) = 0, \qquad g''(x_1) = \frac{10^6}{99}, \qquad g'(x_2) = \frac{128}{25}, \qquad g''(x_2) = -\frac{64}{5}. \tag{3.3}$$

3.2. **A disk in the lower half-plane.** Put $w_j = A(x_j) = g(x_j) - ic$ for $j = 1, 2$, and let

$$b = g(x_2) - 10i, \qquad r = \frac{99}{10}, \qquad D = \{w \in \mathbb{C} : |w - b| < r\}.$$

The closed disk $\overline{D}$ lies in $\{\operatorname{Im} w \le -1/10\}$. From $3 < \pi < 4$ we get $2.97 < c < 3.96$, so $s := 10 - c$ satisfies $6.04 < s < 7.03$. Next,

$$d := g(x_1) - g(x_2) = 1 - 4p + q \log \frac{3p}{1-p} = \frac{24}{25} - \frac{99}{100} \log 33,$$

and $3.49 < \log 33 < 3.50$ gives $|d| < 3$. Since $w_2 - b = is$ and $w_1 - b = d + is$, we have $|w_2 - b| < r$ and $|w_1 - b|^2 < 9 + 49.43 < r^2$, so $w_1, w_2 \in D$. Put

$$\Delta = r^2 - s^2 > 98.01 - 49.4209 = 48.5891.$$

Then

$$\frac{2r}{\Delta} < \frac{19.8}{48.5891} < \frac{5}{12}, \qquad \rho := \frac{r\,|w_1 - w_2|}{|r^2 - (w_1 - b)\overline{(w_2 - b)}|} = \frac{r|d|}{|\Delta + ids|} \le \frac{r|d|}{\Delta} < \frac{29.7}{48.5891} < \frac{2}{3}. \tag{3.4}$$

The number ρ is the pseudo-hyperbolic distance between w_1 and w_2 in D.

3.3. **A local inverse.**

Lemma 3.1. *Let $U = \{1/w : w \in \overline{D}\}$, a compact subset of $\mathbb{C}^+$. There are $\delta > 0$ and $\varepsilon_1 > 0$ such that for $0 < \varepsilon < \varepsilon_1$ there is an analytic map H_ε from the δ-neighborhood U_δ of U into $\mathbb{C}^+$ with $F_{\lambda_\varepsilon}(H_\varepsilon(u)) = u$ for $u \in U_\delta$, and $\sup_{U_\delta} |H_\varepsilon(u) - u| \to 0$ as $\varepsilon \to 0$. The function*

$$R_\varepsilon(w) = H_\varepsilon(1/w) - 1/w$$

is analytic on a neighborhood of $\overline{D}$, satisfies $\operatorname{Im} R_\varepsilon < 0$ on $\overline{D}$, and $\sup_{\overline{D}} |R_\varepsilon| \to 0$ as $\varepsilon \to 0$.

Proof. For $z \in \mathbb{C}^+$, $G_{\lambda_\varepsilon}(z) = \int (z - \varepsilon s)^{-1}\, \mu(ds) \to 1/z$ as $\varepsilon \to 0$ by dominated convergence, since $|z - \varepsilon s|^{-1} \le (\operatorname{Im} z)^{-1}$. The functions G_{λ_ε} are uniformly bounded on compact subsets of $\mathbb{C}^+$, so by Vitali's theorem the convergence is locally uniform together with all derivatives. Hence $F_{\lambda_\varepsilon} \to \mathrm{id}$ locally uniformly on $\mathbb{C}^+$ with all derivatives.

The convex hull of U is a compact subset of $\mathbb{C}^+$. Choose $\delta > 0$ such that the open 4δ-neighborhood V of this convex hull has compact closure in $\mathbb{C}^+$. The set V is convex. Choose ε_1 such that $|F'_{\lambda_\varepsilon} - 1| < 1/2$ and $|F_{\lambda_\varepsilon} - \mathrm{id}| < \delta$ on V for $\varepsilon < \varepsilon_1$. For $v_1 \ne v_2$ in V,

$$F_{\lambda_\varepsilon}(v_2) - F_{\lambda_\varepsilon}(v_1) = (v_2 - v_1) \int_0^1 F'_{\lambda_\varepsilon}\big(v_1 + \theta(v_2 - v_1)\big)\, d\theta,$$

and the integral has real part greater than $1/2$, so F_{λ_ε} is injective on V. Its inverse is analytic on the open set $F_{\lambda_\varepsilon}(V)$. If $u \in U_\delta$, the circle $|v - u| = 2\delta$ lies in V, and on it $|(F_{\lambda_\varepsilon}(v) - u) - (v - u)| < \delta < |v - u|$. By Rouché's theorem $F_{\lambda_\varepsilon} - u$ has a zero in $|v - u| < 2\delta$, so $u \in F_{\lambda_\varepsilon}(V)$. Let H_ε be the inverse restricted to U_δ. Then $|H_\varepsilon(u) - u| = |H_\varepsilon(u) - F_{\lambda_\varepsilon}(H_\varepsilon(u))| \le \sup_V |\mathrm{id} - F_{\lambda_\varepsilon}| \to 0$.

The map $w \mapsto 1/w$ sends a neighborhood of $\overline{D}$ into U_δ, so R_ε is analytic there and tends to 0 uniformly on $\overline{D}$. For $w \in \overline{D}$ put $v = H_\varepsilon(1/w) \in \mathbb{C}^+$. Then $F_{\lambda_\varepsilon}(v) = 1/w$, and Lemma 2.1 gives $\operatorname{Im}(1/w) > \operatorname{Im} v$, that is, $\operatorname{Im} R_\varepsilon(w) < 0$. □

3.4. **Schwarz–Pick estimates.** For $0 < \varepsilon < \varepsilon_1$ let

$$a_\varepsilon = \operatorname{Re} R_\varepsilon(w_2), \qquad t_\varepsilon = -\operatorname{Im} R_\varepsilon(w_2) > 0, \qquad B_\varepsilon = R_\varepsilon - a_\varepsilon, \qquad K_\varepsilon = -\operatorname{Im} B_\varepsilon.$$

Then B_ε is analytic on a neighborhood of $\overline{D}$, maps D into $\mathbb{C}^-$, $B_\varepsilon(w_2) = -it_\varepsilon$, and $t_\varepsilon \to 0$ as $\varepsilon \to 0$. We write $\partial_1 K_\varepsilon$ for the derivative of K_ε in the real direction; since B_ε is analytic, $\partial_1 K_\varepsilon = -\operatorname{Im} B'_\varepsilon$ and $|\partial_1 K_\varepsilon| \le |B'_\varepsilon|$.

Lemma 3.2. *For every compact set $Q \subset D$ and every integer $k \ge 0$ there is a constant $C(Q,k)$, independent of ε, such that $\sup_Q |B_\varepsilon^{(k)}| \le C(Q,k)\, t_\varepsilon$. Moreover*

$$K_\varepsilon(w_1) > \tfrac{1}{5}\, t_\varepsilon, \qquad K_\varepsilon(w_2) = t_\varepsilon, \qquad |\partial_1 K_\varepsilon(w_2)| \le |B'_\varepsilon(w_2)| < \tfrac{5}{12}\, t_\varepsilon.$$

Proof. Write $t = t_\varepsilon$. The map $\zeta \mapsto (\zeta + it)/(\zeta - it)$ sends $\mathbb{C}^-$ into the unit disk, so $\phi = (B_\varepsilon + it)/(B_\varepsilon - it)$ maps D into the unit disk and $\phi(w_2) = 0$. By the Schwarz–Pick lemma, transported to D by $w \mapsto (w-b)/r$,

$$|\phi(w)| \le \frac{r\,|w - w_2|}{\bigl|r^2 - (w-b)\overline{(w_2 - b)}\bigr|}, \qquad |\phi'(w_2)| \le \frac{r}{r^2 - |w_2 - b|^2} = \frac{r}{\Delta}. \tag{3.5}$$

Solving for B_ε gives $B_\varepsilon = -it(1+\phi)/(1-\phi)$. On a compact set $Q' \subset D$ the right side of the first inequality in (3.5) is at most some $\rho_{Q'} < 1$, so $|B_\varepsilon| \le t(1+\rho_{Q'})/(1-\rho_{Q'})$ on Q'. Applying this to a compact neighborhood Q' of Q in D and using Cauchy's estimates gives the bounds on the derivatives. Next,

$$K_\varepsilon = t\,\operatorname{Re}\frac{1+\phi}{1-\phi} = t\,\frac{1-|\phi|^2}{|1-\phi|^2} \ge t\,\frac{1-|\phi|}{1+|\phi|}.$$

At w_1 we have $|\phi(w_1)| \le \rho < 2/3$ by (3.4), hence $K_\varepsilon(w_1) > t\,(1/3)/(5/3) = t/5$. Finally $B'_\varepsilon = -2it\,\phi'/(1-\phi)^2$, so $|B'_\varepsilon(w_2)| = 2t|\phi'(w_2)| \le 2rt/\Delta < 5t/12$ by (3.5) and (3.4). □

3.5. **The density of the convolution near x_1 and x_2.** Let $\tilde\lambda_\varepsilon$ be the translate of λ_ε by $-a_\varepsilon$, and let $\sigma_\varepsilon = \tilde\lambda_\varepsilon \boxplus \kappa$. Then $F_{\tilde\lambda_\varepsilon}(v) = F_{\lambda_\varepsilon}(v + a_\varepsilon)$.

Fix $j \in \{1,2\}$. Since $A(x_j) = w_j \in D$, we may choose $\eta > 0$ such that $N_j = \{z : |z - x_j| \le 2\eta\} \subset N$ and the compact set $Q = \{w : \operatorname{dist}(w, A(N_j)) \le \eta\}$ lies in D. Let $C_0 = \max_{N_j}(|A'| + |A''|)$ and $C_1 = C(Q,0) + C(Q,1)$ from Lemma 3.2. Let $N_j^0 = \{|z - x_j| < \eta\}$.

Lemma 3.3. *There is $\varepsilon_2 \in (0, \varepsilon_1)$ such that for $0 < \varepsilon < \varepsilon_2$ the following hold.*

1. *For each $z \in N_j^0$ the equation $W = A\bigl(z - B_\varepsilon(W)\bigr)$ has a unique solution $W_\varepsilon(z)$ with $|W_\varepsilon(z) - A(z)| \le C_0 C_1 t_\varepsilon$, and W_ε is analytic on N_j^0.*
2. *$G_{\sigma_\varepsilon} = W_\varepsilon$ on $N_j^0 \cap \mathbb{C}^+$. Consequently, on the interval $J_j = (x_j - \eta, x_j + \eta)$ the measure σ_ε is absolutely continuous with the real-analytic density $f_\varepsilon(x) = -\pi^{-1} \operatorname{Im} W_\varepsilon(x)$.*

Proof. Write $t = t_\varepsilon$ and choose ε_2 so small that $C_0 C_1 t \le \eta$, $C_1 t \le \eta$ and $C_0 C_1 t < 1/2$ for $\varepsilon < \varepsilon_2$. Fix $z \in N_j^0$ and let $\mathcal{B}$ be the closed disk of radius $C_0 C_1 t$ about $A(z)$; it is convex and contained in Q. For $W \in \mathcal{B}$, $|B_\varepsilon(W)| \le C_1 t \le \eta$, so $z - B_\varepsilon(W) \in N_j$ and

$$\bigl|A(z - B_\varepsilon(W)) - A(z)\bigr| \le C_0 C_1 t.$$

Thus $\Phi_z(W) = A(z - B_\varepsilon(W))$ maps $\mathcal{B}$ into itself, and $|\Phi_z(W) - \Phi_z(W')| \le C_0 C_1 t\,|W - W'|$ on $\mathcal{B}$. By the contraction principle Φ_z has a unique fixed point $W_\varepsilon(z) \in \mathcal{B}$. The iterates $\Phi_z^n(A(z))$ are analytic in z and converge uniformly on N_j^0, so W_ε is analytic. This proves (1).

Let $z \in N_j^0 \cap \mathbb{C}^+$, $W = W_\varepsilon(z) \in Q \subset D$ and $u = 1/W \in U$. Put

$$\omega_1 = H_\varepsilon(u) - a_\varepsilon, \qquad \omega_2 = z - B_\varepsilon(W).$$

Then $\omega_1 \in \mathbb{C}^+$ because $H_\varepsilon(u) \in \mathbb{C}^+$, and $\operatorname{Im}\omega_2 = \operatorname{Im} z + K_\varepsilon(W) > 0$. We have $F_{\tilde\lambda_\varepsilon}(\omega_1) = F_{\lambda_\varepsilon}(H_\varepsilon(u)) = u$. Since $|\omega_2 - z| \le \eta$, the point ω_2 lies in $N_j \cap \mathbb{C}^+$, where $A = G_\kappa$; hence $F_\kappa(\omega_2) = 1/A(z - B_\varepsilon(W)) = 1/W = u$. Finally $H_\varepsilon(u) = u + R_\varepsilon(W)$, so

$$\omega_1 + \omega_2 = u + R_\varepsilon(W) - a_\varepsilon + z - B_\varepsilon(W) = z + u.$$

By Lemma 2.3, $u = F_{\sigma_\varepsilon}(z)$, that is, $G_{\sigma_\varepsilon}(z) = W_\varepsilon(z)$. Since W_ε is analytic on N_j^0, the Stieltjes inversion formula gives the last assertion. □

3.6. **Two derivatives of opposite sign.** Let $E_\varepsilon(z) = W_\varepsilon(z) - A(z) + A'(z)\,B_\varepsilon(A(z))$ for $z \in N_j^0$. With $t = t_\varepsilon$, Taylor's formula for A on N_j and Lemma 3.2 on the convex set $\mathcal{B}$ give

$$W_\varepsilon(z) = A(z) - A'(z)B_\varepsilon(W_\varepsilon(z)) + O(t^2) = A(z) - A'(z)B_\varepsilon(A(z)) + O(t^2),$$

because $|B_\varepsilon(W_\varepsilon(z))| \le C_1 t$ and $|B_\varepsilon(W_\varepsilon(z)) - B_\varepsilon(A(z))| \le C_1 t \cdot C_0 C_1 t$. The implied constants depend only on C_0 and C_1. Hence $|E_\varepsilon| \le C_2 t^2$ on N_j^0 with C_2 independent of ε, and Cauchy's estimate gives $|E_\varepsilon'(x_j)| \le C_2 t^2/\eta$.

For real $x \in J_j$, (3.1) gives $\operatorname{Im} A(x) = -c$ and $\operatorname{Im}(A'(x)B_\varepsilon(A(x))) = -g'(x)K_\varepsilon(A(x))$. Therefore

$$f_\varepsilon(x) = \frac{1}{\pi}\Big(c - g'(x)\,K_\varepsilon(g(x) - ic) - \operatorname{Im} E_\varepsilon(x)\Big),$$

and since $\frac{d}{dx}K_\varepsilon(g(x) - ic) = g'(x)\,\partial_1 K_\varepsilon(g(x) - ic)$,

$$f_\varepsilon'(x) = -\frac{1}{\pi}\Big(g''(x)\,K_\varepsilon(g(x) - ic) + g'(x)^2\,\partial_1 K_\varepsilon(g(x) - ic)\Big) - \frac{1}{\pi}\operatorname{Im} E_\varepsilon'(x). \tag{3.6}$$

At x_1, (3.3) and Lemma 3.2 give

$$f_\varepsilon'(x_1) \le -\frac{1}{\pi}\cdot\frac{10^6}{99}\cdot\frac{t}{5} + \frac{C_2 t^2}{\pi\eta} = -\frac{200000}{99\pi}\,t + O(t^2).$$

At x_2 they give

$$f_\varepsilon'(x_2) \ge \frac{1}{\pi}\Big(\frac{64}{5} - \Big(\frac{128}{25}\Big)^2\frac{5}{12}\Big)t - \frac{C_2 t^2}{\pi\eta} = \frac{704}{375\pi}\,t + O(t^2).$$

Since $t_\varepsilon \to 0$, there is $\varepsilon_0 > 0$ such that $f_\varepsilon'(x_1) < 0 < f_\varepsilon'(x_2)$ for $0 < \varepsilon < \varepsilon_0$.

3.7. **Conclusion.** Let $0 < \varepsilon < \varepsilon_0$ and suppose that σ_ε is unimodal, with a mode m and a function h as in (1.1). Since σ_ε has no atoms in $J_1 \cup J_2$, we have $h = f_\varepsilon$ almost everywhere on $J_1 \cup J_2$. If $m > x_1$, then h is nondecreasing on an interval around x_1, and by Lemma 2.4 so is f_ε, which contradicts $f_\varepsilon'(x_1) < 0$. Hence $m \le x_1 < x_2$, and h is nonincreasing on an interval around x_2; by Lemma 2.4 so is f_ε, which contradicts $f_\varepsilon'(x_2) > 0$. So σ_ε is not unimodal.

Now let $L > 1/\varepsilon_0$ and $\varepsilon = 1/L$. Then $D_\varepsilon \nu_L = \kappa$, so $D_\varepsilon(\mu \boxplus \nu_L) = \lambda_\varepsilon \boxplus \kappa$, and σ_ε is the translate of this measure by $-a_\varepsilon$. Since positive dilations and translations preserve unimodality, $\mu \boxplus \nu_L$ is not unimodal. This proves Theorem 1.1 with $L_0 = 1/\varepsilon_0$. □

4. One-sided stable kernels

Let $0 < \alpha < 1$ and let S_t be as in (1.2). Replacing t by t/γ we may assume $\gamma = 1$.

Lemma 4.1. *Let $t > 0$.*

1. *S_t has a bounded continuous density g_t on $\mathbb{R}$, and $g_t = 0$ on $(-\infty, 0]$.*
2. *There is $b_t \in \mathbb{R}$ such that g_t is nonincreasing on (b_t, ∞).*
3. *For every $M \in \mathbb{R}$, g_t does not vanish identically on (M, ∞).*

Proof. (1) The function $\lambda \mapsto \int e^{-\lambda x} S_t(dx)$ is continuous on $\{\operatorname{Re}\lambda \ge 0\}$ and analytic on $\{\operatorname{Re}\lambda > 0\}$, and it equals $\exp(-t\lambda^\alpha)$ on $(0, \infty)$, with the principal branch of λ^α. Hence the two functions agree on $\{\operatorname{Re}\lambda \ge 0\}$, and the characteristic function of S_t is $\theta \mapsto \exp(-t(-i\theta)^\alpha)$. Its modulus is $\exp(-t|\theta|^\alpha \cos(\pi\alpha/2))$, which is integrable because $\cos(\pi\alpha/2) > 0$. By Fourier inversion S_t has a bounded continuous density g_t. It vanishes on $(-\infty, 0)$ because S_t is carried by $[0, \infty)$, and $g_t(0) = 0$ by continuity.

(2) Stable distributions are selfdecomposable, so S_t is unimodal by Yamazato's theorem [9]. Since S_t has no atoms, (1.1) holds with a function h that is nonincreasing on (b_t, ∞), where b_t is a mode, and $h = g_t$ almost everywhere. Lemma 2.4 shows that g_t is nonincreasing on (b_t, ∞).

(3) Suppose $g_t = 0$ on (M, ∞), where we may take $M > 0$. Then $\Phi(\lambda) = \int_0^M e^{-\lambda x} g_t(x)\,dx$ is an entire function equal to $\exp(-t\lambda^\alpha)$ for $\lambda > 0$, hence on $\mathbb{C} \setminus (-\infty, 0]$. For $y > 0$, the limits of $\exp(-t\lambda^\alpha)$ as $\lambda \to -y$ from $\mathbb{C}^+$ and from $\mathbb{C}^-$ are $\exp(-ty^\alpha e^{i\pi\alpha})$ and $\exp(-ty^\alpha e^{-i\pi\alpha})$. Their ratio is $\exp(-2ity^\alpha \sin \pi\alpha)$, which is not 1 when $0 < 2ty^\alpha \sin \pi\alpha < 2\pi$. This contradicts the continuity of Φ at $-y$ for small $y > 0$. □

Proposition 4.2. *Let $(p_n)_{n\ge1}$ be positive numbers with $\sum_n p_n = 1$, and let $\mu = \sum_{n\ge1} p_n \delta_{-2^n}$. Then $\mu * S_t$ is not unimodal for any $t > 0$.*

Proof. Fix $t > 0$. Since g_t is bounded, the series

$$q_t(x) = \sum_{n\ge1} p_n\, g_t(x + 2^n)$$

converges uniformly on $\mathbb{R}$, so q_t is continuous, and by Tonelli's theorem it is a density of $\mu * S_t$. In particular $\mu * S_t$ has no atoms.

Let n be such that $2^{n-1} > b_t$, and let $I_n = (-2^n + b_t,\, -2^{n-1})$, a nonempty interval. Let $x \in I_n$. If $k < n$, then $x + 2^k \le x + 2^{n-1} < 0$ and $g_t(x + 2^k) = 0$. If $k \ge n$, then $x + 2^k \ge x + 2^n > b_t$. By Lemma 4.1(2), every term of q_t is a nonincreasing function of x on I_n, and so is q_t.

Suppose that $\mu * S_t$ is unimodal with mode m. Then q_t agrees almost everywhere on $(-\infty, m)$ with a nondecreasing function, and by Lemma 2.4 it is nondecreasing on $(-\infty, m)$. Choose n_0 with $2^{n_0-1} > b_t$ and $-2^{n_0-1} \le m$. For $n \ge n_0$ we have $I_n \subset (-\infty, m)$, so q_t is constant on I_n. For $u < v$ in I_n,

$$0 = q_t(u) - q_t(v) = \sum_{k\ge n} p_k\big(g_t(u + 2^k) - g_t(v + 2^k)\big),$$

and every term is nonnegative, so every term vanishes. Taking $k = n$ shows that g_t is constant on $(b_t, 2^{n-1})$. As $n \ge n_0$ is arbitrary, g_t is constant on (b_t, ∞), and since g_t is integrable the constant is 0. This contradicts Lemma 4.1(3). □

Proof of Theorem 1.2. Let $c_n = 2^{-n}/(1 + f(-2^n))$, $Z = \sum_n c_n \in (0, 1]$ and $p_n = c_n/Z$. Then $\int f\,d\mu = Z^{-1} \sum_n 2^{-n} f(-2^n)/(1 + f(-2^n)) \le Z^{-1}$, and Proposition 4.2 applies. □

Proof of Corollary 1.3. For $r > 0$, $\int |x|^r\,d\mu_* = Z^{-1} \sum_{n\ge1} 2^{-n^2+rn}$. For $n \ge 2r$ the n-th term is at most $2^{-n^2/2}$, so the series converges. Proposition 4.2 applies for every $\alpha \in (0, 1)$, and L_t is the case $\alpha = 1/2$, $\gamma = \sqrt{2}$ of (1.2). □

Example 4.3. For the Lévy kernel L_t every step is explicit. The density $\ell_t(x) = t(2\pi)^{-1/2} x^{-3/2} e^{-t^2/(2x)}$, $x > 0$, satisfies

$$\ell_t'(x) = \frac{t}{\sqrt{2\pi}}\, x^{-7/2} e^{-t^2/(2x)} \Big(\frac{t^2}{2} - \frac{3x}{2}\Big),$$

so ℓ_t is strictly decreasing on $(t^2/3, \infty)$ and positive on $(0, \infty)$. With $b_t = t^2/3$, the measure $\mu_* * L_t$ has a density that is strictly decreasing on each interval $(-2^n + t^2/3,\, -2^{n-1})$, and these intervals move to $-\infty$.

Remark 4.4. For classical Brownian motion with initial distribution μ, Hasebe and Ueda proved that $\int e^{\epsilon x^2}\,d\mu < \infty$ for some $\epsilon > 0$ implies unimodality of $\mu * N(0, t)$ for large t [5, Theorem 1.3]. Theorem 1.2 shows that for one-sided stable kernels of index below 1 no integrability condition, however strong, has this effect. The obstruction comes from the direction of the support of the kernel, not from the tails of μ.

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The arguments of Section 3 and of Proposition 4.2 in the case of the measure μ_* and the index $1/2$ were produced by OpenAI's GPT-6.1 model, run through the Codex agent. Anthropic's Claude Opus 5.5 checked both arguments independently. The same model then re-derived every step, extended the second result to all indices in $(0, 1)$ and to arbitrary integrability conditions, and wrote the manuscript. These checks are not peer review. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Data availability

No data were used.

The code and data for this paper are at `https://github.com/yspennstate/free-strong-unimodality`. The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [10].

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